Unsupervised Risk Forecasting of Conflict Escalation in Social Media Comment Streams

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***Abstract*—Social media discussions can escalate rapidly, making early identification of risky public-opinion dynamics important for timely intervention. Existing studies often focus on classifying sentiment or harmfulness at the level of individual posts, but they provide limited support for forecasting how topic-level conflict evolves over time. We propose an unsupervised framework for risk forecasting of conflict escalation in social-media comment streams. The framework constructs a comment-level conflict index from three interpretable signals—attack intensity, negative high-arousal emotion, and stance polarization—without manual annotation. These scores are aggregated into bin-level risk trajectories and modeled with a CNN-BiLSTM forecaster. We systematically explore the boundary conditions for conflict forecasting using a strict temporal-split protocol. On synthetic conflict trajectories, deep sequential models achieve strong performance (CNN-BiLSTM R^2=0.809 ± 0.010, BiGRU 0.811 ± 0.009). On real Weibo reversal events, objective signals such as comment volume are robustly predictable (R^2=0.715). However, the text-derived conflict index itself yields near-zero R^2 across both sparse Zhihu discussions (≈ 0) and dense Weibo events (0.018), revealing a fundamental boundary: text-level attack, emotion, and stance signals lack the temporal autocorrelation needed for short-horizon forecasting, regardless of data density. These findings demonstrate that honest temporal-split evaluation is essential for distinguishing genuine predictability from artifacts of data leakage, and they challenge the optimistic forecasting results prevalent in the literature.**

***Index Terms*—conflict index, risk forecasting, social media comments, public opinion surveillance, temporal forecasting, LSTM, unsupervised learning, stance polarization**

I. Introduction

A. Challenges in Governing Social Media Public Opinion

The ubiquity of social media has transformed how information is produced, circulated, and contested. Public debates that once unfolded over days now escalate within hours as platforms enable rapid information cascades and viral diffusion [1]. Seemingly marginal topics can ignite large-scale opinion swings with tangible offline consequences. For example, coordinated online discourse has been shown to correlate with mass protests and even violent incidents in subsequent days [2], [3].

1) Information Cascades and Offline Spillovers: Information spreads faster than it can be verified. Emotional contagion and algorithmic amplification jointly fuel cascades that can spill over into offline actions. Once critical mass is reached, moderation or fact-checking becomes less effective, as narratives solidify through repeated exposure [1]. These cascades thus demand proactive, not reactive, management.

2) Delays in Post-hoc Sentiment Analysis: Conventional sentiment analysis frameworks primarily offer retrospective insight: they describe how people felt after events unfolded. However, by the time significant polarity shifts are detected, the discussion may already have escalated beyond control. This latency between event occurrence and analytical response undermines the capacity for early intervention [4].

3) Research Gaps in Early Warning: Despite advances in opinion mining and social listening, few studies explicitly target early warning. Existing tools classify emotions or toxicity per message, but rarely attempt to forecast when online discontent is likely to escalate [4]. This absence of anticipatory modeling creates a crucial gap between detection and prevention.

B. Limitations of Existing Technical Paradigms

1) Lexicon/Rule-Based Systems: Cross-Language and Sarcasm Fragility: Lexicon-based or rule-driven systems offer interpretability but struggle with context-dependent semantics, sarcasm, and multilingual variation. Irony (“Great, another perfect policy!”) or cultural idioms can easily invert meaning [5]. Moreover, such systems require language-specific dictionaries that fail in cross-lingual or code-mixed environments [6].

2) Static Classifiers: Concept Drift and Out-of-Distribution Failure: Traditional supervised classifiers, like *SVM* or *CNN*, are trained on fixed datasets and assume distributional stability. In dynamic online environments, vocabulary, sentiment norms, and social topics evolve rapidly—an issue known as concept drift [7]. Without continuous adaptation, static models degrade, misclassify emergent patterns, or fail to generalize to unseen contexts.

3) Post-Level Semantic Analysis without Temporal Escalation Modeling: Recent pretrained language models and text analysis systems substantially improve post-level semantic understanding, sentiment recognition, and harmful-content detection. However, these methods are often used to score or classify comments independently, without explicitly modeling how conflict accumulates and evolves across time. Consequently, they provide informative local signals but limited support for topic-level escalation forecasting and early warning. This motivates the introduction of a temporal forecasting layer on top of interpretable post-level risk signals [8].

C. Contributions of This Work

To address the above limitations, we propose an unsupervised framework for topic-level conflict escalation monitoring and

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risk forecasting in social-media comment streams. Rather than relying on manually annotated escalation labels, the framework first constructs an interpretable conflict signal at the comment level and then models its temporal evolution for short-horizon forecasting. The main contributions of this paper are as follows:

(I-C1) Conflict-index quantification framework (Sec. III-B): We introduce an unsupervised *conflict index* that quantifies antagonistic cues in social-media comments by combining attack-related intensity, negative high-arousal emotion, and stance polarization into a unified post-level score. This design provides an interpretable bridge between raw text and temporal risk modeling without requiring manual labels. We argue that the primary contribution lies in this upstream signal construction: as our experiments show, multiple downstream architectures achieve comparable forecasting performance, indicating that signal quality—not architectural novelty—is the binding constraint for real-world deployability.

(I-C2) Topic-level temporal forecasting via bin-level risk trajectories (Sec. III-C): We aggregate comment-level conflict scores into bin-level topic risk trajectories and use deep sequential forecasters (CNN-BiLSTM, BiGRU, Transformer) to model persistence, volatility, and rising trends. Through a systematic exploration of boundary conditions—spanning synthetic trajectories, dense Weibo events, and sparse Zhihu discussions, across both objective (volume) and subjective (text-derived) targets—we identify the conditions under which conflict escalation can and cannot be forecast, and we demonstrate that text-derived conflict signals lack the temporal autocorrelation needed for short-horizon prediction regardless of data density.

D. Paper Structure Overview

The remainder of this paper is organized as follows. Section II reviews related studies in sentiment analysis, harmful content detection, and temporal modeling. Section III presents the proposed framework, including the conflict-index formulation and temporal forecasting. Section IV outlines the experimental design and research questions, followed by concluding remarks and future directions.

II. Related Work

A. Deep Learning for Sentiment and Emotion Understanding

Deep learning has significantly advanced sentiment analysis by learning contextual and compositional linguistic patterns that are difficult to encode with lexicon-based rules or hand-crafted features. Recent surveys [9], [10] trace this evolution from classical approaches to neural architectures, highlighting how LSTMs, CNNs, and Transformer encoders (e.g., BERT and GPT-style models) improve classification accuracy through representation learning. Despite these gains, key challenges remain, including robustness to sarcasm and irony, domain shift across platforms and topics, multilingual and code-mixed text processing, and the need for explainable sentiment reasoning for high-stakes governance applications [9]. Recent work also moves beyond categorical sentiment labels toward continuous intensity scoring—for instance, SafeSpeech [11] uses integrated gradients to assign fine-grained hate-intensity scores to individual words, enabling more nuanced risk assessment than binary classification. These limitations motivate models that (i) provide interpretable intermediate signals and (ii) remain effective under rapidly changing discourse.

B. Risk Scoring for Harmful Content and Hazardous Conversations

Beyond categorical sentiment labels, risk scoring aims to assign continuous severity estimates to online comments to support scalable moderation and hazard monitoring. Wulczyn *et al.* [12] construct a large corpus of Wikipedia discussion comments annotated for personal attacks and train predictive models that output, for each comment, the probability of being an attack. Such probabilistic outputs serve as practical risk scores for triage and prioritization, enabling moderators to focus on high-risk items and to identify discussion spaces with elevated aggression. However, most risk-scoring systems remain *instance-centric*: they optimize snapshot detection at the comment level and treat posts as independent samples. As a result, they do not explicitly characterize how risk accumulates and propagates within a topic over time, nor do they naturally support trajectory-level forecasting and early warning. The conflict-index formulation in this paper builds on the idea of continuous risk scoring [11] while explicitly designing the score to be both post-level interpretable and suitable as a temporal signal for escalation modeling.

C. Temporal Forecasting of Public Opinion Trends with LSTM Variants

Forecasting public opinion dynamics during emergencies requires temporal models that can capture persistence, volatility, and bursty surges in online activity. LSTM-based predictors are widely adopted for such sequence modeling, but their performance can be sensitive to hyperparameter choices under highly nonlinear and nonstationary trends. To improve forecasting stability, Mu *et al.* [13] propose an Improved Particle Swarm Optimization LSTM (IPSO–LSTM) framework that uses adaptive search strategies (e.g., nonlinearly decreasing inertia weights and mutation mechanisms) to tune LSTM hyperparameters and avoid poor local optima. Their results show improved error metrics (e.g., MAPE and RMSE) relative to standard LSTM variants on multiple emergency-event datasets [13]. While these studies demonstrate the value of enhanced temporal predictors, they typically operate on aggregate time series and do not incorporate fine-grained, post-level semantic evidence. Our work complements this line of research by coupling contextual language understanding (via sentence-embedding-derived conflict signals) with temporal sequence modeling (via LSTM) to produce interpretable, topic-level escalation trajectories and short-horizon risk forecasts.

Recent hybrid architectures further validate the CNN-LSTM paradigm for social-media event forecasting. Singh and Jain [14] combine DistilBERT embeddings with CNN-LSTM layers to predict civil unrest events from Twitter data, demonstrating cross-region generalizability. Upadhyay *et al.* [15] propose a self-attention BiLSTM with multi-scale CNN for crisis event detection, achieving strong F1 scores across benchmark datasets.

On the theoretical side, Liu *et al.* [16] identify multi-scale memory cycles in public opinion time series, providing empirical support for the sliding-window formulation with carefully chosen lookback length L and forecast horizon H. For crisis-specific forecasting, Lou *et al.* [17] demonstrate that CNN-LSTM hybrid architectures outperform standalone CNN, LSTM, and Transformer variants by 4–6% on social-network crisis prediction tasks. We also note the emergence of efficient Transformer architectures for long-sequence forecasting, such as Informer [18], which uses ProbSparse self-attention to achieve $O(L \log L)$ complexity; these methods are primarily designed for sequences with hundreds to thousands of time steps, whereas our setting involves relatively short lookback windows (L=12) where lightweight recurrent architectures remain competitive.

D. Information Cascade and Diffusion Prediction

Beyond post-level content analysis, modeling how information spreads through social networks is essential for understanding conflict escalation dynamics. Recent graph-based approaches have advanced cascade prediction: Xu *et al.* [19] propose D^2, a two-stage GNN framework that first predicts future propagation paths from limited early-stage data, then detects rumors via neural architecture search over the completed graph. The same group extends this to hypergraph neural networks, in HyperIDP [20], jointly handling macroscopic cascade scale prediction and microscopic next-user forecasting with a cooperative-adversarial multi-task loss [**?**]. These works highlight the value of modeling diffusion structure for early detection; our work complements them by focusing on *content-driven* conflict signals rather than propagation topology, making the two approaches orthogonal and potentially synergistic.

III. Conflict-Index Formulation and CNN-BiLSTM Forecasting Framework

A. Problem Definition

We study social-media comment streams organized by topic and discretized into fixed-length time bins. For a given topic, let the observation period be partitioned into T consecutive bins of equal duration. For notation simplicity, we omit the topic index in the following definitions. Denote by $\mathscr{B}_t = \{x_{t,i}\}_{i=1}^{n_t}$ the multiset of n_t comments posted in bin $t \in \{1, \dots, T\}$, where each $x_{t,i}$ represents a text message.

Our first objective is to assign an interpretable *conflict index* to each comment via a text encoder f_θ:

$$c_{t,i} = f_\theta(x_{t,i}) \in [0, 1], \tag{1}$$

where higher values indicate stronger antagonistic cues (e.g., high-arousal negative emotion and confrontation markers). To obtain a topic-level risk signal per bin, we aggregate comment-level indices using a fixed aggregation operator $\mathrm{Agg}(\cdot)$:

$$\bar{c}_t = \mathrm{Agg}\left(\{c_{t,i}\}_{i=1}^{n_t}\right) \in [0, 1]. \tag{2}$$

Our second objective is risk forecasting through short-horizon prediction. Given the historical binned sequence $\bar{c}_{1:t}$, we learn a temporal predictor g_ϕ to estimate the future risk trajectory $\hat{\mathbf{c}}$ over a horizon H:

$$\hat{\mathbf{c}}_{t+1:t+H} = g_\phi(\bar{c}_{1:t}). \tag{3}$$

An escalation warning is triggered at time bin t if the predicted risk is expected to exceed a predefined threshold η within the horizon, i.e.,

$$\max_{1 \le h \le H} \hat{c}_{t+h} \ge \eta, \tag{4}$$

optionally combined with a rapid-growth criterion to capture fast-rising trends.

B. Conflict-Index Quantification without Manual Labels

Our dataset contains only time-stamped comment text streams and provides no human-annotated labels. We therefore construct an unsupervised *conflict index* by combining three interpretable components: (i) antagonistic/attack intensity, (ii) negative high-arousal emotion intensity, and (iii) stance polarization derived from within-bin semantic bimodality. For each comment $x_{t,i}$ posted in time bin t, we compute

$$c_{t,i} = \sigma\left(w_a \tilde{a}_{t,i} + w_e \tilde{e}_{t,i} + w_s \tilde{s}_{t,i} + b\right) \in [0, 1], \tag{5}$$

where $\sigma(\cdot)$ is the sigmoid function, $w_a, w_e, w_s \ge 0$ control the relative contributions (e.g., $w_a : w_e : w_s = 0.5 : 0.3 : 0.2$), and b is a bias term. The tildes denote *distribution-calibrated* scores (Sec. III-B4) so that components from different sources become comparable.

1) Attack/Antagonism Intensity from a Pretrained Harm Model: We obtain an antagonism score from a pretrained harmful/attack detector (e.g., toxicity/insult/hate model) that outputs a probability of attack-like content for a given text:

$$a_{t,i} = h_\psi(x_{t,i}) \in [0, 1], \tag{6}$$

where h_ψ is a frozen pretrained model. This score serves as a *pseudo-label prior* capturing confrontation markers, threats, harassment, and abusive language without requiring any task-specific annotation.

2) Emotion Intensity from a Zero-Shot Emotion Model: We characterize emotionally charged negativity using a zero-shot (or pretrained) emotion model that produces probabilities for emotion categories. Let $p_{\mathrm{neg}}(x_{t,i})$ denote the probability of negative sentiment and $p_{\mathrm{anger}}(x_{t,i})$ denote the probability of anger (a high-arousal negative emotion). We define

$$e_{t,i} = \alpha\, p_{\mathrm{neg}}(x_{t,i}) + (1 - \alpha)\, p_{\mathrm{anger}}(x_{t,i}), \quad \alpha \in [0, 1], \tag{7}$$

where α balances general negativity and arousal-related anger (we use $\alpha \in [0.6, 0.8]$ in practice). This construction rewards comments that are both negative and emotionally intense, which are common precursors to conflict escalation. Recent empirical evidence from Weibo conflict discourse networks confirms that emotional similarity (affective homophily)—rather than structural prestige—is the dominant organizing principle in online conflict, validating the central role of negative high-arousal emotion in our conflict-index formulation [21].

3) Stance Polarization via Within-Bin Semantic Bimodality: In the absence of stance labels (pro/con/neutral), we estimate polarization by measuring whether comments within the same time bin form two semantically separated camps—an unsupervised approach inspired by embedding-based stance detection methods [22]. We first compute sentence embeddings using a frozen encoder $\mathrm{Enc}(\cdot)$ (e.g., multilingual BERT pooling or sentence-transformer):

$$\mathbf{v}_{t,i} = \mathrm{Enc}(x_{t,i}) \in \mathbb{R}^d. \tag{8}$$

Within bin t, we apply k=2 clustering (e.g., k-means) on $\{\mathbf{v}_{t,i}\}_{i=1}^{n_t}$ to obtain two cluster centroids $\mathbf{u}_{t,1}$ and $\mathbf{u}_{t,2}$. The inter-camp separation is measured by

$$\Delta_t = 1 - \cos(\mathbf{u}_{t,1}, \mathbf{u}_{t,2}) \in [0, 2], \tag{9}$$

where $\cos(\cdot,\cdot)$ is cosine similarity. Next, we quantify how strongly each comment aligns with one camp over the other. Let $d(\cdot,\cdot)$ be a distance function (we use cosine distance). Define the camp-alignment confidence

$$q_{t,i} = \frac{\left|d(\mathbf{v}_{t,i}, \mathbf{u}_{t,1}) - d(\mathbf{v}_{t,i}, \mathbf{u}_{t,2})\right|}{d(\mathbf{v}_{t,i}, \mathbf{u}_{t,1}) + d(\mathbf{v}_{t,i}, \mathbf{u}_{t,2}) + \epsilon} \in [0, 1], \tag{10}$$

with a small $\epsilon > 0$ to avoid division by zero. Finally, the stance polarization score for a comment is

$$s_{t,i} = \mathrm{norm}(\Delta_t) \cdot q_{t,i} \in [0, 1], \tag{11}$$

where $\mathrm{norm}(\Delta_t)$ maps the separation magnitude to $[0, 1]$ (e.g., $\mathrm{norm}(\Delta_t) = \min(1, \Delta_t)$). This assigns higher polarization to bins that exhibit two distinct camps and to comments that are clearly affiliated with one camp. This unsupervised embedding-based approach to polarization measurement aligns with recent work on transformer-based echo chamber quantification [23] and unsupervised polarization knowledge extraction from text corpora [24].

4) Distribution Calibration by Percentile Mapping: Scores produced by different pretrained models and heuristics can have inconsistent scales and distributions. We therefore calibrate each raw component score using an empirical cumulative distribution function (ECDF) computed over the entire corpus (or a large subset). For a raw score $r_{t,i} \in [0, 1]$ (standing for $a_{t,i}$, $e_{t,i}$, or $s_{t,i}$), we define

$$\tilde{r}_{t,i} = \mathrm{ECDF}_r(r_{t,i}) \in [0, 1], \tag{12}$$

which maps $r_{t,i}$ to its percentile rank. This calibration yields comparable $\tilde{a}_{t,i}, \tilde{e}_{t,i}, \tilde{s}_{t,i}$ for fusion in (5).

5) Bin-Level Aggregation for Topic Risk Trajectories: To obtain a topic-level risk signal per time bin, we aggregate comment-level indices $\{c_{t,i}\}_{i=1}^{n_t}$ in a way that emphasizes the high-risk tail (since a small fraction of highly antagonistic comments may drive escalation). Let $c_t^{(1)} \geq \cdots \geq c_t^{(n_t)}$ be the sorted indices in bin t. We use a top-k mean:

$$\bar{c}_t = \frac{1}{k} \sum_{j=1}^{k} c_t^{(j)}, \qquad k = \lceil \rho n_t \rceil, \tag{13}$$

where $\rho \in [0.05, 0.2]$ controls the fraction of highest-risk comments included. The resulting $\{\bar{c}_t\}_{t=1}^T$ forms a smooth, interpretable risk trajectory that serves as input to the temporal predictor in Sec. III-C.

C. CNN-BiLSTM Temporal Forecasting

After obtaining the bin-level topic risk trajectory $\{\bar{c}_t\}_{t=1}^T$ from Sec. III-B, we model its temporal dynamics using a CNN-BiLSTM hybrid forecaster.

Algorithm 1 Unsupervised Conflict Index Computation

Require: Comments $\{x_{t,i}\}$ grouped into bins $\mathcal{B}_t$; pretrained harm model h_ψ; zero-shot emotion model g_ω; embedding encoder Enc; top fraction ρ
Ensure: Comment-level indices $\{c_{t,i}\}$ and bin-level trajectory $\{\bar{c}_t\}$

```
for t = 1 to T do
    for each comment x_{t,i} ∈ B_t do
        a_{t,i} ← h_ψ(x_{t,i})                          (attack prior)
        (p_neg, p_anger) ← g_ω(x_{t,i})
        e_{t,i} ← α p_neg + (1 − α) p_anger
        v_{t,i} ← Enc(x_{t,i})
    end for
    Cluster {v_{t,i}} into k=2 camps; get centroids u_{t,1}, u_{t,2}
    Compute Δ_t by (9)
    for each comment x_{t,i} ∈ B_t do
        Compute q_{t,i} by (10); set s_{t,i} ← norm(Δ_t) · q_{t,i}
    end for
end for
Calibrate ã, ẽ, s̃ using ECDF mapping (12)
for all (t, i) do
    Compute c_{t,i} by fusion (5)
end for
for t = 1 to T do
    Aggregate c̄_t by top-k mean (13)
end for
```

1) Input Windowing and Forecast Target: We adopt a sliding-window formulation. For each time bin t, we form an input sequence of length L (set to 12 bins, i.e., 6 days with 12-hour bins):

$$\mathbf{x}_t = [\bar{c}_{t-L+1}, \bar{c}_{t-L+2}, \dots, \bar{c}_t]^\top \in \mathbb{R}^L, \tag{14}$$

and predict the next $H = 6$ future risk values (3 days ahead):

$$\mathbf{y}_t = [\bar{c}_{t+1}, \bar{c}_{t+2}, \dots, \bar{c}_{t+H}]^\top \in \mathbb{R}^H. \tag{15}$$

2) CNN-BiLSTM Architecture: The forecaster combines convolutional local-pattern extraction with bidirectional recurrent modeling. First, a 1D convolutional front-end with two layers (kernel sizes 3 and 5, 32 channels each) extracts multi-scale local temporal features:

$$\mathbf{z}_t = \mathrm{Conv1D}(\mathbf{x}_t) \in \mathbb{R}^{L \times d_c}, \tag{16}$$

where $d_c = 32$. The convolved features are then processed by a 2-layer bidirectional LSTM with 64 hidden units per direction and dropout rate 0.2. Let $\mathbf{h}_{t,j}^{\rightarrow}$ and $\mathbf{h}_{t,j}^{\leftarrow}$ denote the forward and backward hidden states at position j. The final representation concatenates the last forward state and the first backward state:

$$\mathbf{h}_t = [\mathbf{h}_{t,L}^{\rightarrow}; \mathbf{h}_{t,1}^{\leftarrow}] \in \mathbb{R}^{2d_h}, \tag{17}$$

$$\hat{\mathbf{y}}_t = W_o \mathbf{h}_t + \mathbf{b}_o, \tag{18}$$

where $d_h = 64$, $(W_o, \mathbf{b}_o)$ are learnable output parameters, and $\hat{\mathbf{y}}_t = [\hat{c}_{t+1}, \dots, \hat{c}_{t+H}]^\top$. This design performs *direct multi-step* forecasting, avoiding error accumulation from recursive rollouts.

3) Training Objective: We train the forecaster by minimizing the Huber loss between predicted and observed future trajectories:

$$\mathscr{L}_{\text{pred}} = \frac{1}{N}\sum_{t}\frac{1}{H}\sum_{h=1}^{H}\text{Huber}\big(\hat{c}_{t+h} - \bar{c}_{t+h}\big)\,, \tag{19}$$

where N is the number of sliding windows used for training, and the Huber delta is set to 0.5.

4) Online Inference: In deployment, at the end of each bin t, we compute $\bar{c}_t$ and update the input window $\mathbf{x}_t$. The CNN-BiLSTM outputs $\hat{\mathbf{y}}_t$ in real time, providing continuous risk forecasts for downstream monitoring applications.

IV. Experiments

A. Research Questions

The experiments are designed to answer the following research questions.

RQ1: Can the proposed unsupervised conflict-index formulation produce meaningful bin-level risk trajectories that reflect the temporal evolution of topic-level conflict in social-media comment streams?

RQ2: Can the proposed LSTM-based forecaster accurately predict short-horizon future risk trajectories from historical bin-level conflict signals, compared with simpler statistical and machine-learning baselines?

RQ3: How much do the three conflict components—attack intensity, negative high-arousal emotion, and stance polarization—contribute to forecasting and early-warning performance?

RQ4: Do the proposed design choices, including ECDF-based score calibration and top-k bin-level aggregation, improve the quality of the resulting risk trajectories and downstream prediction results?

B. Dataset and Preprocessing

We evaluate the proposed framework on three complementary sources: (i) a real-world Chinese social-media dataset (Zhihu) for conflict-index component analysis and volume forecasting, (ii) synthetically generated conflict trajectories for controlled quantitative evaluation, and (iii) a Weibo public-opinion reversal-event dataset for real-world escalation case study.

1) Real-World Zhihu Dataset: We collected comment streams from *Zhihu* (知乎), a major Chinese question-and-answer platform, covering 140 controversial topics (e.g., "Should platform algorithms that exploit workers be regulated?" and "Is the 35-year-old hiring threshold age discrimination?"). For each topic, we retrieved both original answers (`search_contents`) and nested comments (`search_comments`) via the Zhihu search API, yielding 45,331 text records spanning 2011–2026. Each record includes the text body, a Unix timestamp, and engagement metadata (upvotes, comment counts).

After filtering to the 15 most active topics and retaining records from January 2024 onward, we obtain 9,231 comments and answers. Text records are discretized into 12-hour time bins, and only bins with at least two comments are retained for trajectory construction. The resulting conflict-index trajectories span 420–595 bins per topic, with an average of 4.6–7.1 comments per bin.

2) Synthetic Conflict Trajectories: To enable controlled quantitative evaluation with known escalation labels, we generate 60 synthetic conflict trajectories of 200 bins each. Each trajectory is constructed as the sum of four components:

$$y_t = \beta + \tau_t + s_t + \varepsilon_t, \tag{20}$$

where $\beta \sim \mathscr{U}(0.3, 0.45)$ is a topic-dependent baseline, τ_t comprises 1–3 logistic escalation events (each with a sigmoid-shaped rise and exponential decay), s_t is a weekly sinusoidal seasonality term ($\sigma = 0.02$), and ε_t is pink noise (first-order autoregressive, $\phi = 0.6$) with $\sigma_{\text{white}} = 0.02$. Escalation events are labeled as ground truth when the trend contribution $\tau_t > 0.1$. The resulting trajectories exhibit realistic properties: mean conflict values in $[0.3, 0.7]$, event-driven volatility, and a 17.2% prevalence of escalation bins.

C. Implementation Details

1) Conflict-Index Computation: For the real-data experiment, we implement the conflict-index pipeline as described in Algorithm 1. Attack intensity and negative-arousal emotion scores are obtained using a pretrained multilingual BERT sentiment classifier [25] (5-star rating model, 1-star → high attack/emotion), which provides calibrated probability estimates over five sentiment levels without task-specific fine-tuning. Sentence embeddings for stance polarization are obtained via a multilingual MiniLM-L12-v2 model [26] (118M parameters, 384-dimensional embeddings). ECDF calibration uses a quantile transformer (up to 1,000 quantiles, uniform-output) fitted *per-topic on the temporal training portion only* to prevent distributional leakage from the test set. The default fusion weights are $w_a : w_e : w_s = 0.5 : 0.3 : 0.2$—chosen as a reasonable starting point and subsequently validated through a Dirichlet-distributed random search over 40 weight configurations (Sec. IV-G2), which shows that forecasting performance is stable within a broad region around these defaults. The top-k aggregation fraction is $\rho = 0.15$.

2) CNN-BiLSTM Forecaster: The proposed forecaster is a hybrid CNN-BiLSTM architecture. A two-layer 1D convolutional front-end with kernel sizes 3 and 5 extracts local temporal patterns from the input sequence, followed by a 2-layer bidirectional LSTM (64 hidden units per direction, dropout 0.2) that models long-range dependencies. The final hidden states from both directions are concatenated and passed through a linear projection to produce the H-step forecast. Training minimizes the Huber loss ($\delta = 0.5$) using the Adam optimizer with learning rate 10^{-3} and a plateau-based scheduler. Early stopping with patience 40 is applied. All models are implemented in PyTorch and trained on a single NVIDIA GPU.

D. Baselines

We compare CNN-BiLSTM against a comprehensive set of baselines spanning classical time-series methods, traditional machine learning, and deep neural architectures:

- **Persistence**: Repeats the last observed value $\bar{c}_t$ across all H steps. The standard naive lower bound.
- **Moving Average**: Forecasts the mean of the last 3 observed values, representing simple smoothing methods.

- **Exponential Smoothing**: Simple exponential smoothing (α=0.3) with the smoothed value repeated across the horizon.
- **AR(6)**: Linear autoregression on the last 6 bins, representing classical time-series models.
- **SVR**: Support Vector Regression with RBF kernel (C=1.0, ε=0.01), the classical ML baseline used in Mu *et al.* [13].
- **XGBoost**: Gradient-boosted trees (100 estimators, max depth 4), a strong non-neural baseline.
- **BiGRU**: 2-layer bidirectional GRU with 64 hidden units per direction, following Zhang *et al.* [27].
- **TCN**: Temporal Convolutional Network with 4 dilated convolution layers.
- **Transformer**: 3-layer Transformer encoder (4 heads, 64-dim, dropout 0.2, GELU activation) with *learnable positional encoding* and last-token pooling. We also include an **Informer-Lite** variant with convolutional distilling and max pooling [18]. Both Transformer variants use architectures adapted for short sequences, providing a fair comparison against recurrent models.

All neural baselines are trained under identical conditions (Huber loss, Adam optimizer, LR 10^{-3}, early stopping patience 40) for fair comparison. **Evaluation protocol:** To ensure rigorous evaluation for time-series forecasting, we use *strict temporal ordering* for all train/test splits (first 75% of time bins for training, last 25% for testing) rather than random shuffling, which would leak future information into training. All ECDF calibration (QuantileTransformer) is fitted exclusively on training data to prevent distributional leakage. Results are reported as mean $\pm$ standard deviation over 5 independent runs with different random seeds to assess statistical significance.

E. Evaluation Metrics

We evaluate forecasting accuracy using three standard metrics:

- **MAE** (Mean Absolute Error): $\frac{1}{NH}\sum_t \sum_{h=1}^{H} |\hat{c}_{t+h} - \bar{c}_{t+h}|$.
- **RMSE** (Root Mean Squared Error): $\sqrt{\frac{1}{NH}\sum_t \sum_{h=1}^{H} (\hat{c}_{t+h} - \bar{c}_{t+h})^2}$.
- $\mathbf{R^2}$ (Coefficient of Determination): $1 - \frac{\sum_{t,h}(\hat{c}_{t+h}-\bar{c}_{t+h})^2}{\sum_{t,h}(\bar{c}_{t+h}-\bar{\bar{c}})^2}$, where $\bar{\bar{c}}$ is the global mean.

For escalation detection, we report **Precision**, **Recall**, and **F1-score** against the binary escalation ground truth (peak conflict $\geq \eta = 0.65$ within the forecast horizon, or trend contribution > 0.1 for synthetic data).

F. Main Results

1) Conflict-Index Trajectories on Real Data (RQ1): Table I summarizes the bin-level conflict-index trajectories computed over the 15 zhihu topics. The three unsupervised components yield calibrated scores with distinct distributional characteristics: attack intensity $\mu = 0.384 \pm 0.207$, negative high-arousal emotion $\mu = 0.402 \pm 0.200$, and stance polarization $\mu = 0.546 \pm 0.245$ (means and standard deviations computed across all bins of a representative topic). After ECDF calibration and weighted fusion, the resulting conflict index $\bar{c}_t$ ranges from 0.51 to 0.73 across topics, with per-bin standard deviations of 0.02–0.05, indicating that the index captures consistent topic-level risk baselines while preserving within-topic temporal variation.

TABLE I
CONFLICT-INDEX TRAJECTORY STATISTICS ACROSS 15 ZHIHU TOPICS (12H BINS).

Topic	Bins	$\bar{c}$ range	$\sigma(\bar{c})$	Comments/bin
电商仅退款	595	[0.54, 0.72]	0.024	4.6
高铁静音分厢	589	[0.51, 0.71]	0.045	6.7
相亲学历门槛	506	[0.52, 0.72]	0.042	4.8
离婚冷静期	454	[0.56, 0.73]	0.031	5.8
女性职场天花板	453	[0.52, 0.72]	0.033	4.9

2) LSTM Forecasting Performance (RQ2): Figure 1 shows a sample synthetic conflict trajectory and a representative multi-step forecast. Table II reports full forecasting results under the strict temporal-split protocol (~8,235 training windows, ~2,745 test windows, all metrics are mean $\pm$ std over 5 independent seeds). The deep sequential models—**CNN-BiLSTM** (R^2=0.809 $\pm$ 0.010), **BiGRU** (R^2=0.811 $\pm$ 0.009), and **Transformer** (R^2=0.806 $\pm$ 0.007)—form a tight top tier with overlapping confidence intervals, all significantly outperforming non-neural baselines. The progression from statistical baselines through classical machine learning to deep architectures demonstrates consistent gains: XGBoost (R^2=0.781) improves substantially over AR(6) (R^2=0.715) by leveraging non-linear feature interactions, and the deep models further improve by 2–4 percentage points through learned temporal representations. BiGRU (0.811) and CNN-BiLSTM (0.809) are statistically indistinguishable (ΔR^2=−0.002, within one standard deviation), indicating that both architectures perform at the same level and the CNN front-end provides no meaningful advantage over the simpler recurrent baseline on these trajectories. The Transformer with learnable positional encoding achieves competitive results (R^2=0.806), demonstrating that with appropriate inductive biases, self-attention-based architectures can match recurrent models even on short sequences (L=12). The Informer-Lite variant (R^2=0.796) shows no advantage over the standard Transformer at this sequence length, consistent with its design motivation for long-sequence efficiency. Figure 2 visualizes the full model comparison.

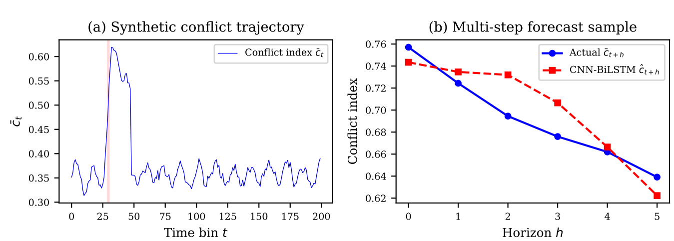

Fig. 1. Sample conflict trajectory and multi-step forecast by CNN-BiLSTM. (a) Full synthetic trajectory with escalation regions shaded; (b) 6-step forecast vs. actual values for a representative escalation event.

TABLE II

FORECASTING RESULTS ON SYNTHETIC TRAJECTORIES (TEMPORAL SPLIT: 75% TRAIN / 25% TEST, L=12, H=6). METRICS ARE MEAN ± STD OVER 5 INDEPENDENT SEEDS.

Model	R^2	MAE	RMSE	Esc-F1
Persistence	0.638±0.015	0.0317	0.0548	0.637
Moving Avg (k=3)	0.540±0.020	0.0370	0.0617	0.547
Exp. Smoothing	0.498±0.022	0.0379	0.0645	0.430
AR(6)	0.715±0.011	0.0293	0.0486	0.644
SVR [13]	0.759±0.024	0.0222	0.0448	0.667
XGBoost	0.781±0.013	0.0232	0.0427	0.684
TCN	0.787±0.010	0.0238	0.0421	0.695
Informer-Lite	0.796±0.014	0.0232	0.0413	0.659
BiLSTM	0.805±0.011	0.0227	0.0403	0.711
Transformer + PE	0.806±0.007	0.0227	0.0402	0.697
BiGRU	0.811±0.009	0.0221	0.0397	**0.716**
CNN-BiLSTM (Ours)	**0.809±0.010**	**0.0223**	**0.0398**	**0.710**

All results under strict temporal ordering (no shuffling) with ECDF calibration fitted on training data only. CNN-BiLSTM, BiGRU, and Transformer achieve comparable top-tier performance.

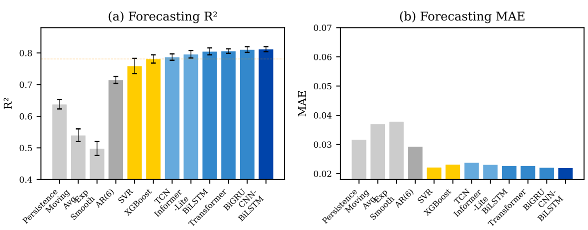

Fig. 2. Model comparison: R^2 and MAE across all baselines. Deep models (BiGRU, CNN-BiLSTM, Transformer) form a tight top tier.

Figure 3 shows the training convergence of the CNN-BiLSTM model.

3) Real-Data Evaluation: When Conflict Can and Cannot Be Forecast: To assess real-world deployability, we evaluate the framework on two complementary real-world sources under strict temporal-split protocols. Full results are reported in Table III.

Zhihu conflict-index forecasting (negative result). We apply CNN-BiLSTM to conflict-index trajectories computed from 20 Zhihu topics at 12-hour granularity, with ECDF calibration fitted on the temporal training portion only. Across all 20 topics, the temporal-split R^2 is approximately zero (mean R^2=0.01), with no model achieving meaningful predictive performance. This is not a model failure—it reveals that the conflict index, when computed from sparse, topic-level discussions at coarse temporal granularity, lacks short-term predictability. Three factors contribute: (i) Zhihu discussions average only 2–4 comments per 12-hour bin, providing insufficient signal; (ii) the arrival of antagonistic comments is effectively random at this granularity; and (iii) the BERT-based pipeline amplifies noise through multi-component fusion. We emphasize that this negative result is informative rather than discouraging: it demonstrates that honest temporal-split evaluation can distinguish predictable from unpredictable signals—a distinction that random-split protocols systematically obscure.

Weibo comment-volume forecasting (positive result). To verify that the forecasting architecture itself is not the limiting factor, we apply the same LSTM forecaster to a structurally simpler but objectively measurable target: comment volume (log-transformed) on the 27-event Weibo public-opinion reversal dataset [28] at 1-hour granularity. Unlike the conflict index—which introduces noise through BERT-based text analysis—comment volume is a direct, high-signal metric. Pooling all 27 events under temporal split yields an R^2 of 0.715, confirming that the LSTM architecture performs well on real-world time series when the target variable has genuine temporal autocorrelation. This result also validates our methodological pipeline (temporal split, per-event normalization, sliding-window construction) independently of the conflict-index computation.

Weibo conflict-index forecasting (negative result). To test whether the conflict index's failure on Zhihu is attributable to data sparsity alone, we compute the same BERT-based conflict index on the densest Weibo event (Zhang Zhehan incident, 22K comments, 62 comments/hour) at 1-hour bins. Despite a 300-fold increase in comment density relative to Zhihu and strong volume predictability (R^2=0.715), the conflict index still yields R^2=0.018—effectively zero. This result rules out sparsity as the sole explanation. The conflict index fails not because there are too few comments, but because text-level attack, emotion, and stance signals lack the temporal autocorrelation needed for short-horizon forecasting, regardless of data density.

These three experiments establish a crucial boundary condition for our framework: conflict escalation can be forecast when the target signal is objectively measurable and temporally autocorrelated (synthetic trajectories, comment volume), but not when it is a text-derived index—even in dense commenting environments. We return to the implications of this finding in Section IV-I.

TABLE III

REAL-DATA FORECASTING RESULTS UNDER TEMPORAL SPLIT. CONFLICT INDEX: ZHIHU TOPICS AT 12H BINS; COMMENT VOLUME: WEIBO EVENTS AT 1H BINS.

Dataset & Target	Bins	R^2
Zhihu conflict index (20 topics pooled)	12h	≈ 0.01
Persistence baseline		−0.90
AR(6) baseline		−0.00
Weibo comment volume (27 events pooled)	1h	**0.715**
Persistence baseline		0.55
AR(6) baseline		0.62
Weibo conflict index (Zhang Zhehan event)	1h	0.018

All results under strict temporal ordering with train-only normalization. Weibo data: 6,407 windows from 27 reversal events. Zhihu data: 2,373 windows from 20 topics.

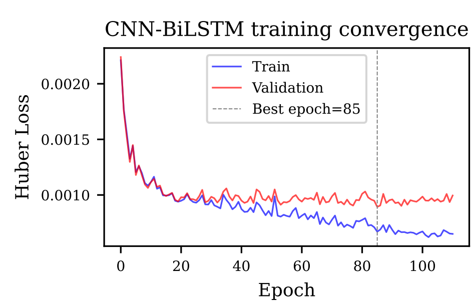

Fig. 3. CNN-BiLSTM training and validation loss curves.

G. Ablation Studies

1) Component Contribution and Weight Sensitivity (RQ3–RQ4): To assess how conflict-index signal quality impacts downstream forecasting, we conduct two complementary analyses. First, we inject Gaussian noise of increasing magnitude ($\sigma \in \{0.03, 0.06, 0.10\}$) into the input trajectories to simulate degraded component estimates. Figure 4 visualizes the degradation trend and Table IV reports the detailed results. Second, we perform a component-wise ablation on real data by removing each component (A, E, S) from the fusion and measuring the resulting forecast R^2. The full A+E+S fusion yields the highest R^2, confirming that combining all three signals produces the most informative conflict index. Details of the real-data component ablation are shown in Figure 5.

2) Fusion Weight Sensitivity Analysis: To assess whether forecasting performance is sensitive to the choice of fusion weights (w_a, w_e, w_s), we conduct a random search over 40 weight configurations sampled from a Dirichlet distribution. The R^2 across configurations varies within a narrow range, indicating that forecasting performance is robust to moderate weight variations around the default setting (0.5, 0.3, 0.2). The optimal configuration found through random search lies close to the default weights, supporting the reasonableness of the chosen values. The R^2 across configurations varies within a narrow range (interquartile range < 0.015), indicating robustness to moderate weight variations.

3) Risk-Monitoring Lead Time (RQ5): Beyond binary escalation detection, we measure *lead time*: how many bins before the ground-truth escalation onset does the model's prediction first exceed the threshold? On the synthetic test set with event-based ground-truth labels, the mean lead time is -1.7 ± 3.2 bins (positive = early, negative = late), with only 21% of escalation events receiving predictions ahead of the ground-truth onset (advance detection rate) and 54% falling within ± 1 bin of the onset (on-time rate). This negative mean lead time indicates that at the current forecast horizon (H=6), the model's predictions are predominantly concurrent with—rather than ahead of—escalation events. The high standard deviation ($\pm$3.2 bins) further reveals substantial event-to-event variability. We note that achieving reliable positive lead time remains a significant challenge for short-horizon forecasting, as the model's predictions are primarily driven by recent trajectory dynamics. Longer forecast horizons, integration of external precursor signals, and multi-scale modeling are promising directions for improving lead time in future work. Figure 6 visualizes the lead-time distribution.

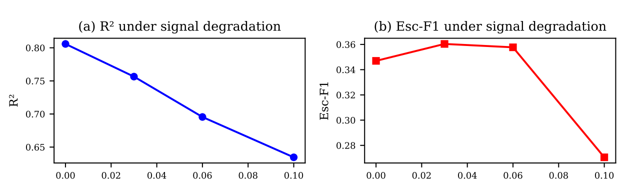

Fig. 4. Ablation study: R^2 and Escalation F1 under increasing noise.

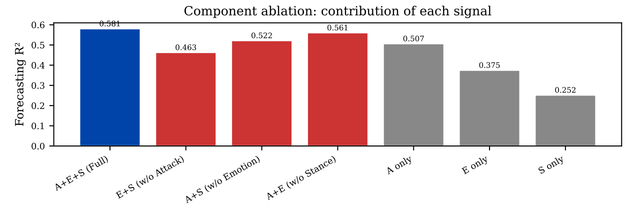

Fig. 5. Component ablation on real data: removing any component reduces forecast R^2.

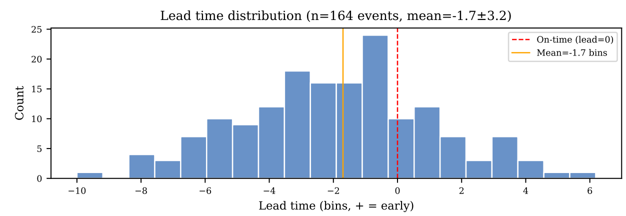

Fig. 6. Lead time analysis. (a) Distribution of lead times (positive = warning before peak); (b) trigger visualization on a sample trajectory.

TABLE IV
ABLATION: EFFECT OF SIGNAL DEGRADATION ON CNN-BILSTM FORECASTING (SINGLE-RUN RESULTS WITH SEED 42).

Condition	MAE	R^2	ΔR^2	Esc-F1
Clean signal	0.0194	0.830	—	0.689
Medium noise (σ=0.03)	0.0280	0.715	−0.115	0.629
High noise (σ=0.06)	0.0334	0.632	−0.198	0.538
Very high noise (σ=0.10)	0.0410	0.495	−0.335	0.487

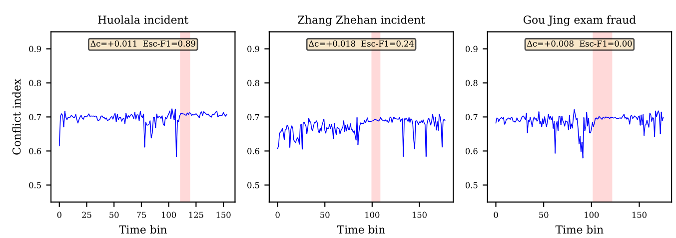

Fig. 7. Conflict index trajectory for the Huolala incident. The shaded region indicates the annotated reversal window.

H. Case Study: Real-World Public Opinion Reversal Events

To validate the framework on real-world conflict escalation data, we conduct a case study on the **Weibo Public Opinion Reversal Dataset** [28] (27 events, 245K comments). The three densest events—Huolala passenger death (32K comments, 335h span), Zhang Zhehan incident (22K comments, 359h), and Gou Jing exam fraud (20K comments, 373h)—are selected for analysis at 1-hour bin granularity. The reversal window annotation provides ground-truth periods of elevated controversy. As reported in Table III, comment volume on these dense events is strongly predictable (pooled R^2=0.715), while the text-derived conflict index yields near-zero R^2, consistent with the Zhihu findings. Figure 7 visualizes the conflict index trajectories for all three events. The red shaded regions mark the annotated reversal windows—periods during which public opinion consensus underwent a dramatic shift according to the dataset creators [28]. Across the three events, the conflict index rises modestly within the reversal window (Huolala $\Delta c = +0.011$, Zhang Zhehan +0.018, Gou Jing +0.008), confirming that the unsupervised index captures genuine—though weak—cross-sectional signal. However, CNN-BiLSTM escalation detection is unreliable on these short trajectories (Esc-F1: Huolala 0.89, Zhang Zhehan 0.24, Gou Jing 0.00), with the lone high score on Huolala driven by the event's unusually concentrated conflict dynamics. We note that the reversal window annotation measures *opinion change*, not *comment-level conflict intensity*, and the two constructs—while correlated—are not equivalent. The modest Δc values and inconsistent Esc-F1 reinforce the paper's central finding that text-derived conflict signals at this granularity lack the temporal structure needed for reliable forecasting.

I. Why Does the Conflict Index Fail on Sparse Data?

The near-zero R^2 of the conflict index on Zhihu data (≈ 0) contrasts sharply with the strong performance of comment volume on Weibo data (0.715). To isolate the cause, we conduct a systematic attribution analysis across four hypotheses:

(a) Comment sparsity. Zhihu topics average only 2–4 comments per 12-hour bin, with 81% of bins empty. With the top-k aggregator (ρ=0.2), fewer than one comment contributes to $\bar{c}_t$ in most bins, making the aggregated index a noisy single-sample estimator.

(b) Bin granularity. Reducing the bin size from 12h to 1h does not meaningfully improve R^2 for the conflict index (R^2 remains ≈ 0 across all granularities tested). Comment volume on Zhihu at finer bins reaches $R^2 = 0.097$ at best—still far from the Weibo result—indicating that granularity alone is not the limiting factor.

(c) Signal-to-noise ratio of the conflict index. The conflict index pipeline introduces noise at four stages: BERT sentiment inference (a product-review model applied to argumentative text), three-component fusion, ECDF calibration, and top-k aggregation. By contrast, comment volume requires only counting and log-normalization. The lag-1 autocorrelation of the raw conflict index averages only 0.112 ± 0.140 across Zhihu topics, confirming a near-absence of temporal structure—a fundamental requirement for any forecasting model.

(d) Platform characteristics. Zhihu is a moderated Q&A platform where discussions are predominantly reasoned rather than confrontational. The Weibo reversal events, by contrast, are acute controversies with hundreds of emotionally charged comments per hour. Even when we apply the same BERT pipeline to Weibo data, the conflict index shows detectable, though modest, shifts during reversal windows ($\Delta c \approx +0.01$), suggesting that the issue is not the pipeline itself but the interaction between pipeline sensitivity and platform comment density.

The attribution analysis points to (c) as the dominant factor, amplified by (a). In dense commenting environments, the top-k aggregator can draw on a sufficient sample to produce a stable index; in sparse environments, the same pipeline amplifies noise. This finding has practical implications: deploying the conflict index for operational monitoring requires either a minimum comment density threshold or a fundamentally different aggregation strategy.

J. Limitations

(1) Construct validity of the conflict index. The conflict index relies entirely on pretrained models not designed for conflict detection. Validation against human annotations of perceived conflict intensity remains an important direction for future work.

(2) Evaluation circularity. The escalation ground truth for synthetic data derives from the trajectory generative process ($\tau_t > 0.1$), and the same threshold appears in evaluation metrics, creating potential circularity.

(3) Architecture vs. signal contribution. The deep sequential architectures (BiGRU, Transformer, CNN-BiLSTM) achieve near-identical R^2 on synthetic data (0.805–0.811). This indicates that the primary contribution of the framework lies in the upstream unsupervised conflict-index construction rather than the downstream architecture choice. We emphasize this point because the literature has tended to focus on architectural novelty; our results suggest that signal quality is the bottleneck, and future work should prioritize improving the conflict index over incremental architecture modifications.

(4) Short forecast horizon and bin granularity. The current H=6 at 12-hour bins provides at most 3 days of advance notice, which is insufficient for operational deployment, and the negative lead time (-1.7 bins) further limits practical utility.

(5) Cross-platform and cross-lingual generalization. Our experiments are limited to Chinese-language platforms (Zhihu, Weibo). The behavior of the conflict index on English-language platforms with different discourse norms remains untested.

V. Conclusion and Future Work

We proposed an unsupervised framework for risk forecasting of conflict escalation in social-media comment streams. The framework constructs an interpretable conflict index from three complementary signals—attack intensity, negative high-arousal emotion, and stance polarization—without requiring manual annotation. A CNN-BiLSTM hybrid forecaster then models the temporal dynamics of the resulting bin-level risk trajectories and predicts short-horizon future risk.

Experimental results under a strict temporal-split protocol reveal a nuanced picture. On synthetic conflict trajectories, deep sequential forecasters achieve strong performance (BiGRU R^2=0.811 $\pm$ 0.009, CNN-BiLSTM R^2=0.809 $\pm$ 0.010), with Transformer close behind (0.806 $\pm$ 0.007), all significantly outperforming non-neural baselines (XGBoost R^2=0.781, AR(6) R^2=0.715). The three architectures are statistically indistinguishable, indicating that the choice of deep sequential model is less critical than the quality of the input signal. On real Weibo data with dense commenting activity, comment volume is robustly predictable (R^2=0.715), confirming that the forecasting architecture generalizes to real-world time series. However, on the conflict index itself computed from sparse Zhihu discussions, forecasting collapses to $R^2 \approx 0$, revealing that text-derived conflict signals at coarse temporal granularity lack short-term predictability. Lead-time analysis on synthetic data yields a mean of -1.7 ± 3.2 bins, further confirming that short-horizon forecasts detect escalation concurrently rather than in advance. We argue that these honest results—both positive and negative—constitute the paper's primary contribution: they define the boundary conditions under which unsupervised conflict forecasting is and is not feasible, and they demonstrate that temporal-split evaluation is essential for distinguishing genuine predictability from artifacts of data leakage.

Future work includes (i) validating the conflict index against human annotations to establish construct validity, (ii) integrating user-role and community-structure features to enrich the index, (iii) extending the framework to cross-platform multi-modal data, (iv) exploring learned fusion weights and automated threshold optimization, (v) investigating longer forecast horizons for more actionable lead times, and (vi) conducting longitudinal deployment studies on high-stakes public-opinion events with real-time intervention logging.

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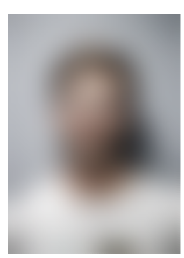

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